

DEPARTMENT OF POSITRON EMISSION TOMOGRAPHY-COMPUTED TOMOGRAPHY

Name	Vipan Kumar Garg	Age/ Sex	68 years/Male
Date	23.03.2026	Referring doctor	Dr Sachin Gupta
MHLI	435714	PET-CT ID	PET/2661/26

Whole Body (¹⁸F-FDG) PET-CT Scan

Brief clinical history: K/c/o Multiple myeloma – post chemotherapy. Now on maintenance Daratumumab. No complaints. PET/CT on follow-up.

Procedure: Approximately 60 minutes after intravenous administration of appx. 11 mCi of [¹⁸F] 2-fluoro-2-deoxy-D-glucose (¹⁸F-FDG), non-contrast enhanced CT was obtained in whole body PET-CT scanner (GE Discovery IQ Gen2) from the vertex to mid thigh. Oral contrast was administered prior to scan. Whole body FDG PET emission scans were acquired over the same anatomical regions.

High resolution images were reconstructed using GE PET adw4.7 and slices were reformatted into transaxial, coronal and sagittal planes. Semi quantitative estimation of FDG uptake was performed by calculating SUVmax values, corrected for dose administered.

Comparison: Previous PET-CT scan done outside dated 12.02.2024

Observations:

Overall bio distribution of FDG is within normal physiological limits.

Head and neck:

No demonstrable abnormal FDG uptake is noted in the supra and infra tentorial brain parenchyma.
(Note: All brain metastases may not be apparent on a PET-CT scan and an MRI may be performed where clinically indicated).

Normal physiologic FDG uptake is seen in the ocular muscles.

Salivary glands demonstrate normal metabolic activity.

Dental implant in-situ is noted with reconstruction artefacts obscuring the regional anatomy.

Mildly increased FDG uptake is noted in bilateral palatine tonsils with tonsilloliths – likely inflammatory.

Normal physiologic FDG uptake is noted in the nasopharynx, oropharynx, hypopharynx and larynx.

Thyroid gland appears unremarkable with no demonstrable abnormal FDG uptake.

No significantly enlarged (short axis diameter) or hypermetabolic cervical lymph nodes are noted.

Thorax:

Mediastinal vascular structures appear unremarkable.

Normal physiological FDG uptake is seen in the myocardium.

Trachea and the main bronchi appear unremarkable.

No evidence of any pleural or pericardial effusions seen.

Subpleural fibroatelectatic changes are noted in lingula and bilateral lung lower lobes. Non FDG avid subcentimetric pleural based nodules (few calcified) are noted in RUL – likely benign.
Few pneumocysts are noted in bilateral lungs.

Faint to FDG avid (SUVmax 5.1) few subcentimetric bilateral lower paratracheal, subaortic, subcarinal and bilateral hilar lymph nodes are noted – likely inflammatory.

Status post hiatus hernia repair.

Abdomen and pelvis:

No evidence of ascites or free fluid seen.

No abnormal FDG uptake is seen in the stomach, small bowel & large bowel loops. Suggested CT enterography/ colonoscopy evaluation if clinically indicated.

Liver measures within normal limits and reveals fairly homogeneous parenchyma with normal physiologic FDG uptake. **Non FDG avid well defined hypodense lesion is noted in segment III of liver (~1.1 x 0.9 cm) – likely benign. Status quo.** Intra hepatic biliary radicles / CBD are not dilated. Portal vein is normal.

Spleen measures within normal limits and reveals fairly homogeneous parenchyma with normal physiologic FDG uptake.

Gall bladder appears fairly well distended with no demonstrable abnormal uptake.

Pancreas appears normal in size and shape with no demonstrable abnormal FDG uptake.

Bilateral adrenal glands appear unremarkable with no abnormal FDG uptake.

Bilateral kidneys appear normal in size and shape with normal physiologic FDG uptake. No evidence of calculus / mass lesion or hydronephrosis is noted. **Mild bilateral perinephric stranding is noted.**

No significantly enlarged (short axis diameter) or hypermetabolic abdominal, retroperitoneal, pelvic or inguinal lymphadenopathy is noted.

Urinary bladder is partially distended and appears grossly normal.

Prostate appears enlarged measuring ~4.7 x 4.1 x 5.0 cm and displays physiological FDG uptake.

Musculoskeletal system:

Normal physiologic FDG uptake is noted in the muscles under view.

Degenerative changes are noted in the visualised spine.

Non FDG avid sclerotic foci are noted in right scapula, left femur and left pubic bone – likely benign (enastosis).

Old healed fractures are noted in right 6th rib laterally (with soft tissue component, SUVmax non avid vs 7.0) and left 8th rib posteriorly. Compared to previous scan, resolution of FDG avidity and decrease in extent of soft tissue component is noted.

Note is made of complete collapse of D1 and D6 vertebral body with sclerosis. Faint to FDG avid right paravertebral soft tissue lesion is noted at D5-D7 vertebral level (SUVmax 4.9 vs 5.8, ~2.5 x 1.5 cm vs 2.6 x 1.8). Compared to previous scan, mild decrease in size and FDG avidity noted.



MAX
Healthcare

25
YEARS OF
SERVICE AND
EXCELLENCE

Non FDG avid lytic skeletal lesions, few with minimal soft tissue component are noted in sternum, multiple cervico-dorsolumbar vertebrae, few ribs, sacrum, bilateral pelvic bones and right femur.

Impression: Scan findings in this case reveal:

- **FDG avid right paravertebral soft tissue component at D5-D7 vertebral level and non FDG avid lytic skeletal lesions (few with minimal soft tissue component, D1 & D6 vertebral collapse and rib fractures) as described – residual disease.**

Compared to previous scan dated 12.02.2024, decrease in extent and FDG avidity pre-existing lesions noted s/o favourable response to therapy.

Please correlate clinically & with other relevant investigations also.

Note: All tumours are not FDG avid and all FDG avid lesions are not malignancies. In absence of metabolically active disease on PET scan, if there are other evidences / strong suspicion to suggest presence of disease then correlation with other investigations & histopathology is advisable.


Dr. Avanthiga.S.S MD
Consultant, Nuclear Medicine

**YEARS OF
SERVICE AND
EXCELLENCE**

Max Super Speciality Hospital, Mohali
(A unit of Hometrail Buildtech Pvt. Ltd.)
Near Civil Hospital, Phase - 6, Mohali,
Punjab - 160 055
For medical service queries or appointments,
call: +91-172 521 2000

www.maxhealthcare.in

Hometrail Buildtech Pvt. Ltd.
Regd. Office: N - 110, Panchsheel Park,
New Delhi - 110 017
T: +91-70424 01894
E: secretarialhbp1@maxhealthcare.com

(CIN: U45400DL2008PTC176962)



H-2013-0181
May 28, 19 - May 27, 22
Since May 28, 2013

